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The dark side of robot usage for hotel employees: An uncertainty management perspective

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ABSTRACT

The increasing usage of service robots in hotels has generated discussions about their positive impacts. However, little research has been done on the adverse aspects of robot usage from the perspective of the employees, and few studies have investigated the antecedents of employee robot risk awareness. This study posits that employees are aware of potential threats posed by robots; they observe the organization's investment in robot development and witness customers' satisfaction with robot usage. People with greater robot risk awareness engage in more withdrawal behaviors at work (employee sample in Study 1) and report lower intentions to work in the hospitality industry (student sample in Study 2). Those two relationships are further contingent upon employees' learning goal orientation. The proposed relationships were validated by a survey-based study (612 frontline employees) and two experiments (205 student participants). Based on the research results, hotels are encouraged to emphasize augmentation but not automation.

1. Introduction

The hospitality industry is known for its large workforce and labor-intensive nature, marked by demanding workloads and long hours (Chen & Wu, 2017). Over the past few decades, driven by technological advancements, hotels have embraced the integration of service robots to improve the work patterns of hotel employees, service quality, and productivity (Xu et al., 2023). It is inevitable that hotels will adopt new technology to support daily operations in the near future. The positive impacts of service robots on organizations and customers have been well documented in academia. At the organizational level, service robot adoption has been found to help organizations create a favorable corporate image and reduce operational costs (Hwang et al., 2023; Pizam et al., 2022). At the customer level, interacting with service robots enhances customers' positive evaluations, hospitality experience, booking intentions, and online rating scores (Filieri et al., 2022; Mariani & Borghi, 2021; Qiu et al., 2020; Romero & Lado, 2021). Customers have also been found to be more tolerant of robot service failures (Yang et al., 2022). Recent studies have argued that overall customer satisfaction is not significantly influenced even when robot performance is unsatisfactory (Borghi & Mariani, 2023).

Despite these benefits, there is an increasing focus in research on the negative implications of service robot adoption from the viewpoint of employees. A recent review by Xu et al. (2023) indicated that service robot adoption has detrimental effects on employees, such as heightened job insecurity, physical and mental fatigue, job burnout, turnover intention, and decreased job satisfaction and work engagement. It has also been argued that this kind of robot adoption in the workplace will generate different emotions among employees, and employees' perceptions are not always positive (He et al., 2023). This kind of negative perception has been called robotics awareness, or robot risk awareness in some recent studies (Li et al., 2019; Yu et al., 2022). This finding is unsurprising, considering that the roles of employees and service robots closely resemble each other for firms that use robots. The current research expands upon and enhances previous studies by proposing that the favorable interactional experience of important others (i.e., organizations and customers) may accelerate employees' negative perceptions. Uncertainty management theory (Lind & Van den Bos, 2002) posits that people have a desire for a sense of security, while uncertainty is often experienced as aversive and alarming. When facing uncertainty, individuals often feel anxious about their capability to control the environment and the quality of outcomes they may receive. Drawing from this insight, we argue that a hotel's service robot adoption strategy creates uncertainty in the workplace, and contextual cues that employees find to be threatening (i.e., high organizational investment in robot development and high customer satisfaction with robot usage)

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strongly stimulate employees' negative reactions. Accordingly, organizational investment in robot development and customer satisfaction with robot usage are expected to positively influence employees' robot risk awareness and withdrawal behaviors at work.

Uncertainty management theory further suggests that as uncertainty can jeopardize one's overall sense of self, people seek buffers or strive to reduce the stress resulting from uncertainty and make uncertainty manageable. Nevertheless, these behaviors depend on individual capabilities (Pierro et al., 2014; Thau et al., 2007; Yam et al., 2023). The uncertainty-reducing function in our research context is reflected in the concept of learning goal orientation. Learning goal orientation refers to an orientation to master a challenging situation by developing competence (VandeWalle, 1997). Employees with greater learning goal orientation tend to view threats as challenge stressors and exhibit more adaptive behaviors (Ma et al., 2021; Peng et al., 2019). As a result, robot risk awareness is less likely to elicit destructive work behaviors. Thus, we propose a moderated mediation model that examines both employees' (Study 1) and students' (Study 2) robot risk awareness as a mediating mechanism and investigates the role of learning goal orientation as a moderator of the relationship between organizational investment in robot development, customer satisfaction with robot usage, and employees' withdrawal behavior (Study 1) as well as students' intentions to work in the hospitality industry (Study 2).

Prior studies on robots have examined employee-robot dyadic interactions only. The present study broadens our understanding of how external pressure may also influence the employees' cognitive and behavioral reactions to robots. This study provides a novel perspective by shifting the role of employees from being the focus to acting as an observer, and argues that "*how employees perceive the way others view robots*" could be as important and significant as employees' own personal perceptions of robots. To achieve this goal, we employ mixed methods: a survey to identify a meaningful effect in the field, paired with an experiment to elucidate the causal relationship.

2. Theoretical background and hypothesis development

2.1. Contextual information: perceived robot risk in the hospitality industry

Service robots, usually defined as "computer programs that perceive, think, and act in order to enhance human productivity and/or engage in social interaction" (Wirtz et al., 2018, p. 909), are now spreading across various hospitality sectors (Prentice et al., 2020). As Borghi and Mariani (2021) observed, service robots have increasingly become a popular feature as noted in customer evaluations of hotel stays rather than a mere novelty. On the positive side, service robots can speed up tasks traditionally performed by hotel service employees, such as concierge, guest check-in, room service, bartending, and chatting (Ivanov et al., 2017; Tung & Law, 2017). However, some recent studies have shown that the use of robots may instill a sense of uncertainty because the rapid development of automation is out of one's control (Seo & Lee, 2021; Tussyadiah et al., 2020). McKnight et al. (2011) argued that even people who trust automation and technology may still be uncertain as to whether the automated robot will fulfill the expected task obligations either intentionally or unintentionally.

Recent literature has addressed concerns about negative perceptions of robotics among hotel employees. The concept of Smart Technology, Artificial Intelligence, Robotics, and Algorithms (STARA) awareness was first introduced by Brougham and Haar (2018), and refers to the extent to which employees feel that their job could be replaced by STARA. This concept has been widely used in many subsequent studies of employees and specific types of new technologies. For example, Kong et al. (2021) focused on the area of artificial intelligence (AI) and examined how AI awareness affects employees' career competency and job burnout. This kind of negative perception may lead to employees' fear of unemployment, or increase their psychological stress (Parvez, Öztüren, et al., 2022; Zhang & Jin, 2023); therefore, Human Resource Management should consider how to eliminate employees' fear of robots and help employees accept robots as team members (Arslan et al., 2022).

In terms of the relationship between hotel employees and robots, Li et al. (2019) applied the concept of robotics awareness and examined the dark side of hotel robots in terms of their negative impact on employees. Similarly, Ivanov et al. (2020) described robot awareness as "fear of automation" and argued that it stems from employees' perceptions of being replaced by robots. Yu et al. (2022) further clarified this kind of robot awareness as "robot risk awareness" to reflect the dark side of employees' perceptions. These studies imply that the dark side originates not only from the rapid development of AI or service robots but also from employees' negative perceptions, as they feel uncertainty about their jobs and careers during this period of rapid robot development. Following Yu et al.'s (2022) work, the term "robot risk awareness" is applied in this study. It is regarded as "the extent to which employees feel their job could be replaced by robots, and it reflects the uncertain situation that is harmful to employees" (Brougham & Haar, 2018; Kong et al., 2021).

According to uncertainty management theory, coping with the various uncertainties people experience in their social relationships is one of the greatest challenges in their lives (Lind & Van den Bos, 2002; Van den Bos & Lind, 2002). The need for predictability and a reduction in uncertainty can be said to have an evolutionary basis (Gudykunst, 1998). As uncertainty poses a challenge to an individual's general sense of self, people strive to develop coping mechanisms to tolerate it or to make it more cognitively manageable (Van den Bos & Lind, 2002). While some articles in the hospitality literature have discussed the dark side of robots in terms of uncertainty (Khaliq et al., 2022; Parvez, Arasli, et al., 2022; Zhong et al., 2020), they have concentrated on the employees' fear of unemployment. Most recent studies have focused on the potential outcomes when employees perceive robot risk awareness (Ding, 2021; Liang et al., 2022), but very few have investigated the underlying reasons why employees generate such risk awareness. Additionally, since most studies have examined only the relationships between employees and robots (Ivanov et al., 2017; Yu et al., 2022), the role of organizations and customers in determining the ultimate performance of robots has been largely overlooked (Chiang & Trimi, 2020). Therefore, it is crucial to consider how organizations and customers perceive the performance of hotel robots.

2.2. Triggers of uncertainty: organizational investment in robot development

Employees' beliefs are often related to how much the organization trusts them, recognizes their contributions, and takes care of their wellbeing. These beliefs are increasingly valued and recognized as key indicators for evaluating perceived organizational investment (Arogundade et al., 2015). Enterprises, especially those based in the service sector, are more likely to adopt new robot technologies, as more investment in machines can both save costs and improve corporate efficiency to promote economic growth (Brougham & Haar, 2018). At present, the hotel industry is paying more attention to investing in robots, and many hotels have even increased their investment in robots annually (Li et al., 2019). Hotel frontline employees are often the first group of employees to be aware of robotics (Yam et al., 2023). While the introduction of robots in hotels may have both positive and negative effects on employee behaviors, employees' feelings toward them are often negative or even destructive (Parvez, Arasli, et al., 2022).

Additionally, resources are limited, and increasing the investment for one thing will inevitably lead to the reduction and neglect of the resources for another thing (Gebauer & Friedli, 2005; Lee & Bruvold, 2003). When an organization increasingly invests in robots, the resources available for employees decrease, which results in a series of uncertain reactions to the organization (Boyd & Holton, 2018; Li et al., 2019), not to mention some of the darker side impacts of robot investment, such as putting millions of jobs at risk and creating mass unemployment (Castillo et al., 2021; Hu & Min, 2023). A significant finding from Brougham and Haar's (2018) study highlights that when organizations proactively explore today's advanced workforce technological solutions, such as artificial intelligence and robotic systems, employees feel uncertain about the new technology and feel unappreciated by the company. In conclusion, if employees perceive increasing investment in advanced technology, they may feel uncertain and experience more negative emotions about robots (Brougham & Haar, 2018). Therefore, the following hypothesis is drawn.

Hypothesis 1. Organizational investment in robot development is positively related to employees' robot risk awareness.

2.3. Triggers of uncertainty: employees' perceptions of customer satisfaction with robot usage

Currently, hotels provide robot services to guests not only internally, but also throughout the entire customer journey to facilitate customers' consumption and experience (Borghi & Mariani, 2021; Li et al., 2019). In other words, robots are part of the service a business provides to its guests and can influence the customer experience to generate positive attitudes (satisfaction) and customer behavior (including revisits, the desire to share their experience with friends, and so on) (Jain et al., 2023; Prentice, Dominique Lopes et al., 2020).

Since employees interact frequently with customers in hotel workplaces, they can perceive customers' reactions when customers encounter robots for the first time (Epaminonda et al., 2020; Yam et al., 2023). This perception will also affect employees' understanding of and behavior toward robots to a large extent (Lu et al., 2020). When service robots can be continuously improved to satisfy customers' demands (Parvez, Öztüren, et al., 2022), Parvez, Öztüren, et al., 2022mployees may fear for their job security, as they cannot satisfy customers in the same way that robots can, thereby generating a stronger sense of uncertainty about their work, which may lead to employees' robot risk awareness accompanied by mental stress (Lu et al., 2022; Yu et al., 2022). Based on the extant literature discussed above, the following hypothesis is proposed.

Hypothesis 2. Employees' perceptions of customer satisfaction with robot usage is positively related to employee's robot risk awareness.

2.4. Employee's withdrawal behaviors at work

The continuous development of modern technology is one of the sources of work stress for employees. Withdrawal behavior is defined as "behavior that causes the psychological or physical distance between the work environment and the employee" (Rosse & Hulin, 1985, p. 325). It includes multiple types of employee behaviors (Liu et al., 2019). For example, psychological withdrawal behaviors include absenteeism, daydreaming, taking care of personal matters while at work, excessive chatting with colleagues, not making efforts to work, letting others do the job, and so on. Physical withdrawal behaviors include early departures from the workplace, prolonged lunch breaks beyond the designated time, and unauthorized removal of work-related materials from the premises. Confrontational behavior includes arguing with coworkers, disobeying supervisors, intentionally spreading rumors or gossip, filing formal complaints, and reporting wrongdoing by others. (Lehman & Simpson, 1992). According to uncertainty management theory, when employees experience uncertainty, their concern for fairness becomes more prominent (Thau et al., 2009). Consequently, when employees perceive a threat to their expertise and a potential erosion of their skills as a result of evolving workplace technologies, they frequently experience high levels of anxiety, sadness and frustration, as well as a pervasive sense of uncertainty, leading to a greater sense of unfairness regarding their own development. When this sense of unfairness reaches a certain level, employees will compromise or withdraw at work (Gabriel & Pessl, 2016).

Furthermore, it is obvious that the total amount of internal investment for an organization is often fixed and limited, and if investment in one part of the organization increases, it usually means that investment in other parts will decrease (Stein, 2003). When employees perceive that the organization is increasing its investment in robots, they will give the organization less positive feedback and engage in withdrawal behaviors (Parvez, Öztüren, et al., 2022). Additionally, it is worth noting that employees' robot risk awareness may play an important mediating role in the employee-organization investment relationship. Recent studies have argued that when employees experience the advanced technology of robots, they may feel insecure and unappreciated because an increase in robot investment means that many middle-skilled and managerial positions will decrease (Dixon et al., 2021). This leads to negative emotions such as fear, which may lead to many negative outcomes such as the intention to quit (Brougham & Haar, 2018). Therefore, based on the above, we propose the following hypothesis.

Hypothesis 3. Robot risk awareness mediates the relationship between organizational investment in robot development and employees' withdrawal behaviors at work.

Similarly, in the hospitality industry, customer satisfaction is an important indicator of the quality of employees' work (Hurley & Estelami, 2007). Thus, hotel employees often try to improve their service quality to gain customers' recognition of their services (Hansemark & Albinsson, 2004). Current research has shown that in terms of employee fear, employees often compete with other colleagues to improve customer satisfaction with their services (Cina, 1989). Moreover, employees may have negative emotions when they perceive that customers are satisfied with the work of their competitors (Prasetio et al., 2019), and have doubts and uncertainties about their own work, resulting in negative behaviors (Hartline & Ferrell, 1996).

When a robot becomes a "new colleague" of the employees and customers show their satisfaction with robot usage, the employees may feel that robots have become "new competitors," and competition between employees is likely to become competition between employees and robots, as most agree that service robots perform some jobs more efficiently than humans (Yam et al., 2023). Such competitions may cause employees to fear robots, which may further affect their withdrawal behaviors. Therefore, the following hypothesis is proposed.

Hypothesis 4. Robot risk awareness mediates the relationship between employees' perceptions of customer satisfaction with robot usage and withdrawal behaviors at work.

2.5. Moderating role of learning goal motivation

The robot literature has further asserted that individual psychological interventions can mitigate the negative consequences caused by robot exposure. For instance, Yam et al. (2023) reported that the positive effects of robot exposure on burnout and workplace incivility via job insecurity are less pronounced among people with high self-affirmation. While most studies have focused on self-affirmation, where people confirm their self-worth and make fewer threat appraisals, the present study focuses on individual adaptability to the changing environment, i.e., the individual's learning goal orientation. Learning goal orientation refers to "a mastery orientation that individuals aim to acquire new skills and improve their own abilities" (Templer et al., 2022, p. 482). Employees who have high learning goal orientation tend to engage in challenging tasks because they are eager to improve their skills, and they often compare their current performance to their past performance (Button et al., 1996). Learning-oriented individuals emphasize developing and seeking to master a new set of skills (Kim & Lee, 2013), perceive their abilities as malleable (Dweck, 1986), and strive to improve task performance (Latham & Seijts, 2016).

According to the achievement goal approach, employees with high learning goal orientation will put in effort and persist in completing complex tasks in the absence of external rewards (Dweck et al., 1995) because challenging work provides them with opportunities to develop skills and knowledge (Morrison & Phelps, 1999; Vandewalle et al., 2019). As a result, employees with high learning goal orientation may view spending time with robots at work as a challenging task rather than a threatening task. In contrast, those with low learning goal orientation may underestimate the value of additional learning opportunities and therefore be less willing to engage in the task (Malik et al., 2019). Therefore, compared to employees with low learning goal orientation, employees with high learning goal orientation are more focused on intrinsically motivated task situations, and their desire to learn new ideas and skills highlights their tendency to persevere effectively in the face of obstacles (Alexander & Van Knippenberg, 2014). In our research context, employees with high learning goal orientation are more capable of coping with robot risk awareness by altering their jobs to achieve a better fit between themselves and the work environment.

It is noteworthy that, for theoretical reasons, learning goal orientation is expected to moderate only the relationship between robot risk awareness and employees' withdrawal behaviors (the second stage of the indirect effect) but not the relationship between uncertainty factors and robot risk awareness (the first stage of the indirect effect). This is because the information that contextual cues convey is an "objective truth." Employees inevitably feel that they are in an unfavorable position when they perceive that service robots are favored by organizations and customers. However, different employees may cope with robot risk awareness differently, and employees' learning-goal orientation is regarded as an individual's ability to cope with this stressor. Cognitive appraisal theory (Lazarus & Folkman, 1984) provides a good theoretical rationale for this reasoning. In the primary appraisal, for a certain work encounter (contextual cues sent by the organization and customers), employees first evaluate whether they have something at stake in the encounter or whether the encounter would result in either beneficial or harmful consequences (robot risk awareness). Then, in the secondary appraisal, employees evaluate whether anything can be done to prevent harm or improve the benefits (learning-goal orientation is an individual coping resource).

Taken together, the dark side effects of the relationship between robot risk awareness and withdrawal behaviors at work are less salient among employees with high learning goal orientation. Combining all the aforementioned hypotheses, we propose the following additional hypotheses.

Hypothesis 5. Employees' learning goal motivation moderates the indirect effect of organizational investment in robot development on withdrawal behaviors at work via robot risk awareness.

Hypothesis 6. Employees' learning goal motivation moderates the indirect effect of employees' perceptions of customer satisfaction with the robot usage on their withdrawal behaviors at work via robot risk awareness.

3. Overview of studies

To test our hypotheses, we conducted two studies with different research designs and samples that collectively investigated the impacts of highly uncertain contextual information about robots on employees' maladaptive emotions and behaviors at work (see Fig. 1). Study 1 was a survey-based study that targeted hotel employees. The survey included measurements of organizational investment in robot development, customer satisfaction with robot usage, robot-induced fear, learning goal orientation, and withdrawal behaviors at work. Participants were required to assess the real situation they observed in the current hotel and to provide their emotional and behavioral reactions. However, our design in Study 1 did not allow us to infer causal relationships among the study variables since all the variables were examined at a single point in time. Therefore, we adopted the experimental-causal-chain approach using upcoming graduates as the sample in Study 2. The full moderated mediation model was examined via two separate experiments. In Study 2A, we manipulated contextual uncertainty factors and measured robot risk awareness. Once this relationship was confirmed, we proceeded to Study 2B, in which we manipulated robot risk awareness and measured learning goal orientation and intention to work in the hospitality industry. In that sense, Study 2 not only allowed us to establish the causal inference for the research model but also enabled us to explore the extent to which organizational openness to robot adoption attracts prospective employees.[1]

4. Study 1: survey-based study

4.1. Method: participants and procedure

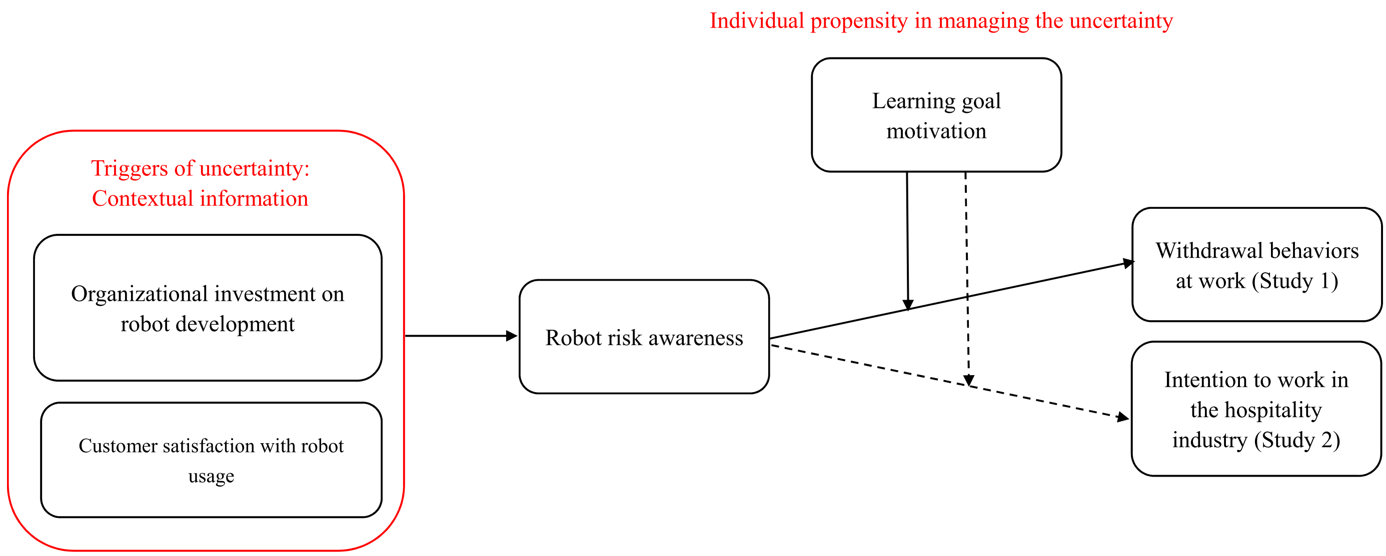
We collected data from hotels in southern China. We focused specifically on hotels located in four cities in the Great Bay Area in China because it has been developed as an international science and technology innovation hub (Li et al., 2019). Accordingly, advanced technologies including service robot adoption, have been introduced in several industries. A growing number of hotels use service robots to attract hotel guests (Borghi & Mariani, 2023). Therefore, the characteristics of these hotels fit our research context well. Next, two research assistants helped to identify eligible participating hotels by visiting online hotel booking websites (e.g., Xiecheng and Qunar). Hotels that had adopted service robots were targeted. We contacted the human resource departments of these selected hotels. With the consent of the hotel representatives, employees who satisfied the following two requirements were further invited to participate in the survey: (1) they had experience working with robots in their daily work, and (2) they had had the opportunity to observe the interactions between customers and service robots. A total of 700 copies of the survey were initially distributed, and 620 were returned. After eliminating those with missing data, 612 responses were considered valid and were used for the subsequent analysis. Among the 612 participants, all held non-management jobs, 51.8% were female, and 66% were between 20 and 40 years of age. In terms of departments, 41.8% worked in the front office, 24.3% worked in housekeeping, and 16.2% worked in food and beverage. On average, they had worked for 3.18 years in their current hotel and had a total of 5.03 years of experience in the hospitality industry.

4.2. Measures

We invited two bilingual individuals to translate all the English measures to Chinese, and followed Brislin's (1970) translation-back-translation practice. Unless otherwise specified, participants were asked to provide answers based on a seven-point scale ranging from 1 (strongly disagree) to 7 (strongly agree).

Organizational investment in robot development. Adapting Cavusoglu et al.'s (2015) three-item scale, we rated participants' perceptions of the extent to which their organization invested resources in developing the robot. While the original items were developed to measure organizational investment in information security control resources, "security" was replaced by "robots" for each item. A sample

[1] An employee sample was used in Study 1 (survey) because we tested their real cognitive and behavioral perceptions at work in the current organization. In contrast, Study 2 had an experimental design and included participants who were randomly assigned to an experimental scenario. We evaluated their subsequent reactions based on the context of the scenario. Therefore, "whether they were a hotel employee" was not a necessary requirement to join the study. However, to ensure participants could fully understand and involve themselves in the scenario, year 4 college students who had hospitality internship experience were chosen as the sample in Study 2. Empirically, it is a common practice to use student samples for experiments to validate the results of a field survey.



Note: The variables and relationships drawn in solid line were tested in Study 1, and those drawn in dotted line were tested in Study 2

Fig. 1. Research model.

item is "I think that our hotel has deployed large resources for robots." Cronbach's alpha for this scale is 0.90.

Employees' perceptions of customer satisfaction with robot usage. Adapting Jia et al.'s (2021) four-item scale, we measured employees' perceptions of customer satisfaction with robot usage in their hotels. While the original items were developed to measure customer satisfaction with service robots, we modified the items by changing "I" to "I feel our customers." A sample item is "I feel our customers were very pleased with the service robot." Cronbach's alpha for this scale is 0.94.

Robot risk awareness. Adapting Li et al.'s (2019) four-item scale, we replaced the term "artificial intelligence" in the original items with "service robots" to better fit the research context. A sample item is "I think my job has a high risk of bowing to automation and will be replaced by service robots." Cronbach's alpha for this scale is 0.90.

Learning goal motivation. Using a five-item measure developed by VandeWalle (1997), participants rated their learning goal orientation. A sample item is "I often look for opportunities to develop new skills and knowledge." Cronbach's alpha for this scale is 0.95.

Withdrawal behaviors at work. We measured employees' withdrawal behaviors at work using twelve items developed by Lehman and Simpson (1992). Participants were required to assess how often they engaged in psychological and physical withdrawal behaviors at work in the past six months, using a seven-point scale ranging from 1 (never) to 7 (very often). A sample item is "I put less effort into my job than I should have." Cronbach's alpha for this scale is 0.98.

4.3. Confirmatory factor analyses

Prior to hypothesis testing, we conducted confirmatory factor analysis to evaluate the convergent validity and discriminant validity of the constructs. As shown in Table 1, the fit of the hypothesized five-factor model was acceptable ($\chi2 = 902.31$, df = 340, CFI = 0.97, TLI = 0.97, RMSEA = 0.05, SRMR = 0.05) and better than that of an alternative measurement model in which all the items were loaded on a general latent factor ($\chi2 = 7413.70$, df = 350, CFI = 0.63, TFI = 0.60, RMSEA = 0.18, SRMR = 0.20; $\Delta\chi2 = 6511.39$, Δdf = 10, $p < 0.05$).

4.4. Results and discussion

Descriptive statistics, reliabilities, and intercorrelations are presented in Table 2. As expected, (a) organizational investment in robot development was positively related to robot risk awareness ($r = 0.67$, $p < 0.01$); (b) employees' perception of customer satisfaction with robot usage was positively related to robot risk awareness ($r = 0.54$, $p < 0.01$); and (c) robot risk awareness was positively related to withdrawal behaviors at work ($r = 0.78$, $p < 0.01$). The correlation results offered the initial support for our hypothesized relationships.

Next, we used Mplus 7.0 to test our hypotheses. The findings from the path analyses are reported in Table 3. We found support for Hypothesis 1 positing that organizational investment in robot development was positively associated with robot risk awareness ($b = 0.56$, $p < 0.05$). Hypothesis 2 predicted employees' perceptions of customer satisfaction with robot usage. As shown in Table 3, we found a significant positive effect of employees' perception of customer satisfaction with robot usage on robot risk awareness ($b = 0.14$, $p < 0.05$), supporting Hypothesis 2. Furthermore, we tested the indirect effects (H3 and H4) and moderated mediation effects (H5 and H6). Bootstrapping with 5000 resamples was employed to construct bias-corrected confidence intervals. Hypothesis 3 and Hypothesis 4 proposed the mediating effects of robot risk awareness. The bootstrap results supported these two hypotheses, as we found that organizational investment in robot usage has a significant indirect effect on withdrawal behaviors at work via robot risk awareness (estimate = 0.53, 95% CI [0.427, 0.638]). Likewise, we also found a significant indirect effect from customer satisfaction with robot usage to withdrawal behaviors at work through robot risk awareness (estimate = 0.13, 95% CI [0.050, 0.212]). Therefore, both Hypothesis 3 and Hypothesis 4 were supported.

The final moderated mediation hypotheses focus on the extent to which learning goal motivation moderates the effect of organizational investment in robot development (H5)/employees' perception of customer satisfaction with robot usage on withdrawal behaviors at work through robot risk awareness. First, we found that robot risk awareness interacted with learning goal motivation in predicting withdrawal behaviors at work ($b = -0.08$, $p = 0.05$). Second, simple slope analysis showed that the positive relationship between robot risk awareness and withdrawal behaviors at work was weaker when employees' learning goal motivation was high (simple slope = 0.80, $p < 0.05$) than when employees' learning goal motivation was low (simple slope = 1.10, $p < 0.05$). We plotted the interaction for the conditional value of learning goal motivation (1 standard deviation above and below the mean)

Table 1
Factor loadings of each item and the test of construct validity of study variables in **Study 1**.

Items	Standardized factor loading	AVE	CR
Organizational investment on robot development		0.75	0.90
I think that our hotel has made large investments in service robots in the past.	0.87**		
I think that robot is among the highest IT spending priorities of our hotel.	0.86**		
I think that our hotel has deployed large resources for service robots.	0.86**		
Employee' s perception of customer satisfaction with robot usage		0.78	0.92
I think the customers are very happy to interact with the service robot.	0.87**		
I think it is convenient for the customers when services provided by service robots.	0.90**		
I think the customers are very pleased with the service robot.	0.89**		
I think, overall, the customers are satisfied with the service robot.	0.88**		
Robot risk awareness		0.70	0.87
I think my job has a high risk of bowing to automation and will be replaced by service robots.	0.82**		
There is a high likelihood that my job in this hotel will become automated.	0.74**		
I am quite pessimistic about my future in this hotel due to employees could be replaced with service robots.	0.89**		
I am uncertain about my future due to many jobs in the hospitality industry are already being replaced by service robots.	0.87**		
Learning goal motivation		0.78	0.95
I am willing to select a challenging work assignment that I can learn a lot from.	0.87**		
I often look for opportunities to develop new skills and knowledge.	0.89**		
I enjoy challenging and difficult tasks at work where I'll learn new skills.	0.89**		
For me, development of my work ability is important enough to take risks.	0.87**		
I prefer to work in situations that require a high level of ability and talent.	0.87**		
Withdrawal behaviors at work		0.78	0.98
Thoughts of being absent.	0.85**		
Chat with co-workers about non-work topics.	0.85**		
Left work station for unnecessary reasons.	0.89**		
Daydreaming.	0.88**		
Spent work time on personal matters.	0.89**		
Put less effort into job than should have.	0.90**		
Thoughts of leaving current job.	0.86**		
Let others do your work.	0.89**		
Left work early without permission.	0.91**		
Taken longer lunch or rest break than allowed.	0.88**		
Taken supplies or equipment without permission.	0.89**		
Fallen asleep at work.	0.90**		

(Fig. 2). Third, we found that organizational investment in robot development had a weaker positive indirect effect on withdrawal behaviors at work via robot risk awareness among employees with high learning goal motivation (indirect effect = 0.45, 95% [0.357, 0.538])

Table 3
Results of hypothesis testing in Mplus in **Study 1**.

Paths	Estimate	Standard Errors	Confidence interval
Direct effects			
Organizational investment on robot development → Robot risk awareness (**H1**)	0.56**	0.05	(0.469, 0.651)
Employee' s perception of customer satisfaction with robot usage → Robot risk awareness (**H2**)	0.14**	0.04	(0.052, 0.224)
Robot risk awareness → Withdrawal behaviors at work	0.95**	0.05	(0.862, 1.039)
Mediating effects			
Organizational investment on robot development → Robot risk awareness → Withdrawal behaviors at work (**H3**)	0.53**	0.05	(0.427, 0.638)
Employee' s perception of customer satisfaction with robot usage → Robot risk awareness → Withdrawal behaviors at work (**H4**)	0.13**	0.04	(0.050, 0.212)
Moderating effect and Simple slop test			
Learning goal motivation → Withdrawal behaviors at work	−0.23**	0.08	(-0.380, −0.078)
Robot risk awareness*Learning goal motivation → Withdrawal behaviors at work	−0.08*	0.04	(-0.154, −0.013)
High learning goal motivation	0.80**	0.05	(0.693, 0.905)
Low learning goal motivation	1.10**	0.12	(0.876, 1.328)
Moderated mediation effects (**H5 & H6**)			
High learning goal motivation: Organizational investment on robot development → Robot risk awareness → Withdrawal behaviors at work	0.45**	0.05	(0.357, 0.538)
Low learning goal motivation: Organizational investment on robot development → Robot risk awareness → Withdrawal behaviors at work	0.62**	0.09	(0.444, 0.789)
Difference	−0.17*	0.09	(-0.318, −0.021)
High learning goal motivation: Employee' s perception of customer satisfaction with robot usage → Robot risk awareness → Withdrawal behaviors at work	0.11**	0.04	(0.039, 0.181)
Low learning goal motivation: Employee' s perception of customer satisfaction with robot usage → Robot risk awareness → Withdrawal behaviors at work	0.15**	0.05	(0.056, 0.248)
Difference	−0.04*	0.03	(-0.083, −0.001)

Note: N = 612; ** 95% confidence level, *90% confidence level.

Table 2
Descriptive statistics and correlations among study variables in **Study 1**.

Variables	Mean	AVE	CR	Standard deviation	1	2	3	4	5
1. Organizational investment on robot development	4.1225	0.75	0.90	1.82456	(0.90)				
2. Employee' s perception of customer satisfaction with robot usage	4.6021	0.78	0.92	1.87481	0.70[a]	(0.94)			
3. Robot risk awareness	3.5915	0.70	0.87	1.80107	0.67[a]	0.54[a]	(0.90)		
4. Learning goal motivation	4.5288	0.78	0.95	1.81293	0.77[a]	0.85[a]	0.58[a]	(0.95)	
5. Withdrawal behaviors at work	3.1743	0.78	0.98	1.87648	0.52[a]	0.35[a]	0.78[a]	0.39[a]	(0.98)

Note: N = 612.
[a] $p < .05$. Diagonal shows the values of Cronbach's alpha.

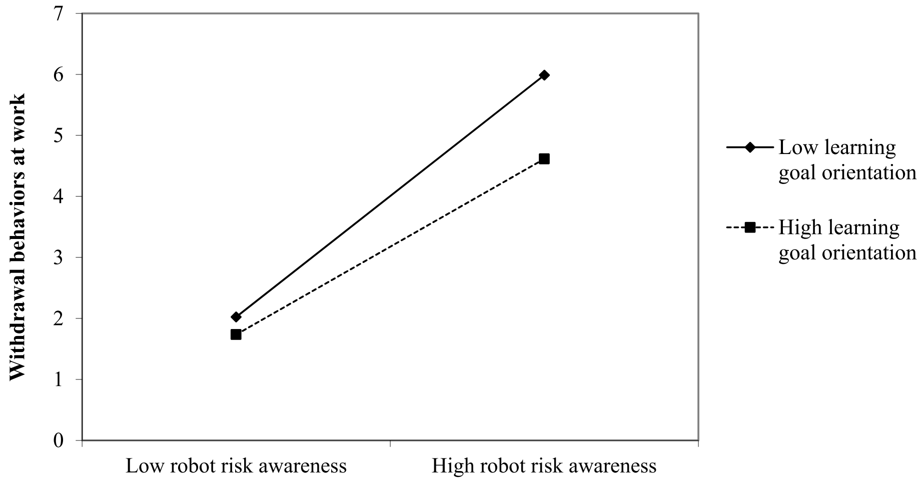


Fig. 2. The moderating effect of learning goal orientation on the relationship between robot risk awareness and withdrawal behaviors at work in **Study 1**.

than among those with low learning goal motivation (indirect effect = 0.62, 95% [0.444, 0.789]). The difference between these two paths was significant (difference = −0.17, 90% CI [−0.318, −0.021]). Similarly, employees' perception of customer satisfaction with robot usage was also found to have a weaker indirect effect on withdrawal behaviors at work via robot risk awareness among the employees with high learning goal motivation (indirect effect = 0.11, 95% [0.039, 0.181]) than among those with low learning goal motivation (indirect effect = 0.15, 95% [0.056, 0.248]). The difference between these two paths was significant (difference = −0.04, 90% CI [−0.083, −0.001]). Taken together, the results supported Hypothesis 5 and Hypothesis 6.[2]

5. Study 2: experiment

Study 1 demonstrated that employees' robot risk awareness is elicited by two contextual factors: organizational investment in robot development and customer satisfaction with robot usage. We also found that employees with a higher level of robot risk awareness are more likely to display more withdrawal behaviors at work. This influential process is heightened among employees with low learning goal motivation. Although the results of Study 1 supported all the hypotheses, (1) the causal nature of the relationships among the study variables could not be confirmed, and (2) the external validity was low. To address these limitations and examine the robustness of the research results, an experimental designed was utilized in Study 2.

To address the aforementioned problems, we adopted the "experimental-causal-chain" approach (Spencer et al., 2005) as a means of breaking down and evaluating the indirect effects within their constituent segments. This approach was employed to offer a stronger evidential basis for establishing causality (Hill et al., 2021). More specifically, in Study 2A, we tested the causal effect of contextual uncertainty factors (i.e., organizational investment in robot development, and the perception of customer satisfaction with robot usage) on robot risk awareness by manipulating contextual uncertainty factors and measuring robot risk awareness. In Study 2B, we tested the causal effect of robot risk awareness on the intention to join the hospitality industry, and whether this effect is moderated by learning goal orientation. We first measured learning goal orientation, manipulated the level of robot risk awareness elicited by contextual uncertainty factors, and measured participants' intentions to work in the hospitality industry. Taken together, these two studies enable us to make causal inferences about our hypotheses.

5.1. Study 2A – antecedent manipulation

5.1.1. Method: participants and procedure

We invited 110 upcoming graduates majoring in hospitality and tourism management at a university located in Macao to participate in this experiment. All of them had at least six months of internship experience and were expected to join the labor market within the next few months. Students voluntarily participated in the experiment by scanning a QR code in a classroom setting. Following the removal of 5 incomplete samples, a total of 105 responses were used in the subsequent analysis. Among the 105 participants in the valid sample, 78.1% were female, and the average age was 20.36 years old (SD = 1.38).

We used the "experimental vignette method," which is an experimental design in which participants are asked to read a carefully constructed and realistic scenario and assess the dependent variables (Aguinis & Bradley, 2014). This method has been widely used in management and psychology research and in studies on human-robot interaction (Choi et al., 2020; Lin & Mattila, 2021; Qiu et al., 2020). Participants were assigned to one of two experimental conditions in a random manner, and in each condition, they were instructed to envision themselves as frontline employees in a fictional hotel that implemented a service robot. We then presented them with a description of the company's and customers' attitudes toward the service robot.

In the high uncertainty condition (N = 51), participants were told that the company's and customers' attitudes toward the service robot were positive. That is, the top management team would continue to invest resources in robot development and increase the number of service robots in the near future. Meanwhile, customers were satisfied with their interactions with the service robot because the service robot saved queuing time and could handle their problems properly. In contrast, in the low uncertainty condition (N = 54), participants were told that the company's and customers' attitudes toward the service robot were negative. On the one hand, the company's future operational strategy would focus on employee training and development. The service robot was utilized to handle trivial and repetitive tasks to reduce the workload and increase the work efficiency of the frontline employees. On the other hand, since the service robot was unable to be customized to suit customers' needs and was not sensitive to customers' feelings and emotions, the service provided by the service robot was unfriendly and

[2] To test the possibility of a moderation effect of learning goal orientation in the first stage, we regressed organizational investment in robot development (X1), customer satisfaction with robot usage (X2), the product term of organizational investment in robot development and learning goal motivation (X1*M), and the product term of customer satisfaction with robot usage and learning goal motivation (X2*M) in terms of their effect on robot risk awareness. The results showed that learning goal orientation failed to significantly moderate both paths.

inconvenient for most customers. Therefore, customers still preferred to interact with employees.

Upon reading the scenario, participants rated their level robot risk awareness using the same scale as in Study 1 ($\alpha = 0.85$; 1 = strongly disagree, 7 = strongly agree). For the manipulation check, participants also completed the measures related to organizational investment in robot development and customer satisfaction with robot usage from Study 1 (for organizational investment in robot development, $\alpha = 0.87$; for customer satisfaction with robot usage, $\alpha = 0.90$; 1 = strongly disagree, 7 = strongly agree).

5.2. Results

As expected, the independent *t*-test results showed that participants in the high uncertainty condition (high organizational investment in robot development and high customer satisfaction with robot usage) reported greater organizational investment in robot development ($M = 5.70$, $SD = 0.89$) and greater perceived customer satisfaction with robot usage ($M = 5.81$, $SD = 0.76$), than did those in the low uncertainty condition (low organizational investment in robot development and low customer satisfaction with robot usage) (organizational investment in robot development $M = 5.22$, $SD = 1.32$, $t(103) = -2.17$, $p < 0.05$; perceived customer satisfaction with robot usage $M = 5.39$, $SD = 1.26$, $t(103) = -2.10$, $p < 0.05$). This indicated that our manipulation was successful.

Furthermore, participants who were in the high uncertainty condition reported a greater level of robot risk awareness ($M = 4.94$, $SD = 1.09$) than did those in the low uncertainty condition ($M = 4.40$, $SD = 1.10$; $t(103) = -2.56$, $p < 0.05$). Therefore, Hypotheses 1 and 2 were supported.

5.3. Study 2B – Mediator manipulation

5.3.1. Method: participants and procedure

To avoid response overlap and obtain an adequate sample size, in Study 2B, we recruited 107 graduate students who were majoring in hospitality and tourism management and had internship experience. The recruitment process was similar to that of Study 2A. Students voluntarily participated in the experiment by scanning a QR code in a classroom setting. Seven participants did not pass the attention check. Among the 100 valid participants, 66% were male. On average, the participants were 24.1 years old ($SD = 2.28$) and had 4.46 months of internship experience ($SD = 4.88$).

In Study 2B, we manipulated robot risk awareness because a key requirement of the experimental-causal-chain design is that the "proposed psychological process as it is measured in Study 1 and as it is manipulated in Study 2 are in fact the same variable" (Spencer et al., 2005, p. 846). Participants first completed a five-item measure of learning goal orientation using the same measure as in Study 1 ($\alpha = 0.81$), after which they were randomly assigned to either a low robot risk awareness condition ($N = 46$) or a high robot risk awareness condition ($N = 54$). We asked all the participants to imagine that they worked in a hotel with a service robot, and further described the performance of the service robot in the hotel. In the high robot risk awareness condition, participants were informed that people believed that service robots can have many positive effects on both organizations and customers, and that the robots could work more efficiently than humans. Hence, they felt threatened and were pessimistic and uncertain about their future career in this hotel. In contrast, in the low robot risk awareness condition, participants were informed that the service robot did not function as well as the organization initially expected. Therefore, employees understood that they were irreplaceable and felt optimistic about their future careers in this hotel.

Following the presentation of the scenario, the participants were asked to provide the rating to gauge their level of intention to work in the hospitality industry using a two-item scale developed by Walsh et al. (2015) ($\alpha = 0.78$, 1 = strongly disagree, 7 = strongly agree). Last, as a check for manipulation, participants filled out the same four-item measure of robot risk awareness employed in Study 1 ($\alpha = 0.86$, 1 = strongly disagree, 7 = strongly agree).

5.4. Results

The results confirmed the success of our manipulation, as the independent *t*-test results indicated that the participants in the high robot risk awareness condition reported a greater level of robot risk awareness ($M = 4.21$, $SD = 1.24$) than did those in the low robot risk awareness condition ($M = 3.02$, $SD = 1.19$; $t(98) = -4.91$, $p < 0.001$).

Since internship experience has a profound impact on career expectations (Wong & Liu, 2010), it was included as a control variable in the subsequent analysis. Having used a causal-chain design and examined first stage and second stage mediating effects in two separate experiments, we could not provide a direct test for Hypotheses 3 and 4—the indirect effects of organizational investment in robot development/employees' perception of customer satisfaction with robot usage on the intention to work in the hospitality industry via robot risk awareness. Nevertheless, we employed the Hayes (2017) PROCESS macro for SPSS (Model 1) to test the moderating effect of learning goal orientation on the relationship between robot risk awareness and the intention to work in the hospitality industry. First, the results showed a negative main effect of the robot risk awareness manipulation on the intention to work in the hospitality industry ($b = -3.79$, $p < 0.05$), and a positive interactive effect ($b = 0.61$, $p < 0.05$). Notably, the main effect of learning goal orientation on the intention to work in the hospitality industry was not significant ($b = -0.57$, $p = 0.23$). Fig. 3 shows the pattern of interaction. A simple slope analysis further revealed that robot risk awareness was negatively related to the intention to work in the hospitality industry only when learning goal orientation was low (simple slope $= -0.84$, $p < 0.05$), but not when learning goal orientation was high (simple slope $= 0.14$, $p = 0.69$). Taken together, Study 2A and 2B provided support for the proposed causal link and successfully replicated the results of Study 1.

6. Discussion

This study developed and tested a research model that pinpoints the dark side of service robot usage from the employee's perspective and investigates the role of employees' robot risk awareness. We found that two situational factors, organizational investment in robot development and customer satisfaction with robot use, significantly induced employees' robot risk awareness, and employees with higher robot risk awareness were more likely to exhibit withdrawal behaviors at work. Moreover, compared to employees with higher learning goal orientation, employees with lower learning goal orientation were more strongly influenced. Therefore, on the basis of previous studies (Cellar et al., 2011; Li et al., 2019; Parvez, Arasli, et al., 2022; Yu et al., 2022), this study took a step further and examined the antecedents of employees' robot risk awareness and the moderating role of learning goal orientation.

Study 2 showed that the proposed model can also be applied to interns who are just starting out in the hospitality industry. Both the organization's investment in robot development and customer satisfaction with robot use can induce interns' robot risk awareness. Interns with higher robot risk awareness are more likely to show less intention to join the hospitality industry. Moreover, compared to interns with lower learning goal orientation, interns with higher learning goal orientation experienced the influence more strongly. Therefore, the relationships proposed by the present research model were verified from different perspectives.

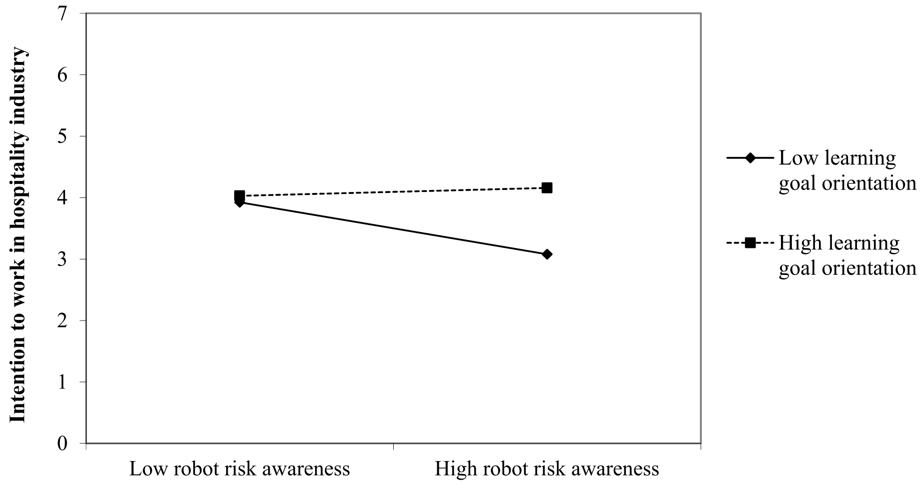


Fig. 3. The moderating effect of learning goal orientation on the relationship between robot risk awareness and intention to work in the hospitality industry in **Study 2.**

6.1. Theoretical implications

This study makes the following three contributions. First, we investigated more possible antecedents of robot risk awareness. Many recent studies have shown that robot risk awareness stems from employees' personal feelings and perceptions. For example, prior research has shown that employees develop their perception of service robots through three things: their previous interactional experiences with service robots, the effort they expect to put into the collaborative process with service robots, and their expectations regarding the possibility of employee–robot collaboration achieving the expected work result (Parvez, Arasli, et al., 2022; Song et al., 2022). However, the mechanism that generates robot risk awareness is complex. The present study considers the feelings and experiences of important others (i.e., organizations and customers) in the same work environment to be determinants of employees' robot risk awareness. Consistent with this notion, social psychologists have noted how individuals are affected by other people (Latané, 1981). In particular, the more people express the same opinion (as the number of sources increases), the greater the impact on an individual's attitudes/behaviors (Spoelma et al., 2023). In our context, the organization's attitude toward and customers' interactional experiences with service robots are expected to have a profound effect on an individual employee's robot risk awareness. When both the organization's and customers' attitudes toward service robots are positive (i.e., high organizational investment in robot development, and high customer satisfaction with robot usage), it exerts the greatest social impact and social pressure on employees, and signals the highest level of uncertainty. This perspective adds to what we previously knew about employees' robot risk awareness.

Our second contribution addresses the role of employees' learning goal orientation in ameliorating the negative impacts of robot risk awareness. Although some research has indicated that individual characteristics may influence robot adoption (Hu & Min, 2023; Xie et al., 2022), employees' self-regulatory strategies (i.e., learning goal orientation) have rarely been studied in the employee–robot collaboration context. Existing research has focused mainly on the moderating effect of employees' demographic factors. For instance, Brougham and Haar (2018) indicated that the negative effect of new technology on individuals' work-related outcomes was more salient among young employees. Similarly, Parvez, Öztüren, et al. (2022) found that entry-level employees react more negatively to the adoption and use of service robots than do their higher-level counterparts. In the present study, we proposed a specific self-regulatory strategy (i.e., learning goal orientation) as enabling employees to better cope with the stress and uncertainty of service robots. Moreover, given that learning goal orientation is malleable (Da Motta Veiga & Turban, 2014), the present study also suggests an effective organizational intervention to counterbalance employees' negative response to service robots.

Third, we advanced uncertainty management theory by applying it to the employee–robot collaboration context. The unified theory of acceptance and use of technology and the uncanny valley theory are two primary theories used to explain the individuals' perceptions of artificial intelligence devices (e.g., a review by Chi et al., 2020). Although the unified theory of acceptance and use of technology examines artificial intelligence adoption from the user perspective (Gursoy et al., 2019) and the uncanny valley theory focuses on the characteristics of artificial intelligence devices (Qiu et al., 2020), uncertainty management theory analyzes this phenomenon from the situational perspective (Andrews et al., 2009). Uncertainty management theory holds that employees are sensitive to contextual information when they are confronted with uncertainty (Loi et al., 2012). Given that service robot adoption presents a certain level of uncertainty in the workplace, employees are especially sensitive to organizational and customer attitudes toward service robots. When organizational and customer attitudes toward service robots are positive, employees' negative cognitive perceptions (i.e., robot risk awareness) and behavioral reactions (i.e., withdrawal behaviors) will increase. The present study demonstrated the additional explanatory power of situational factors on employees' robot risk awareness.

6.2. Practical implications

Given the widespread use of service robots in travel and hospitality companies, managing employee–robot relationships is an important issue (Huang & Rust, 2021). This study provides a practical guide that can help tourism managers to innovate and to ease relationships by adjusting employees' focus on robots. First, our findings suggest that tourism and hotel company managers should recognize the importance of organizational investment in robotics and customer satisfaction with robotic services, and must proceed with caution when their companies attempt to employ service robots for work. Although robots can improve the working efficiency of the enterprise and increase customer satisfaction (Grewal et al., 2021), excessive organizational investment and high customer satisfaction may cause employees to worry, and could lead to many negative effects and withdrawal behaviors. Therefore, companies should balance the investment ratio between robots and employees, and reward their employees in a timely manner when customers feel satisfied with employee service. This approach can effectively alleviate the tension between employees and robots. Additionally, companies should provide annual or quarterly announcements of the company's investment in robots so that employees will have a clearer picture of the company's planning in this regard. Another way to relieve employee stress is through continuous recognition from both the company and customers so that employees understand that the workplace of the future requires employees and robots to coexist, and that employees will not be replaced.

Second, our results showed that employees' robot risk awareness can increase their withdrawal behaviors at work. Managers should improve communication with employees to understand their thoughts and concerns, especially in regard to working with robots, so that managers can address employees in a timely manner when they show negative feelings and a crisis level of robot risk awareness. Finally, employees' learning goal motivation strategies should be consistent with their motivation characteristics. Compared with employees with a lower degree of learning goal orientation, employees with a higher degree of learning goal orientation are more likely to appreciate service robots and feel less uncertainty in regard to robots. This means that subdividing employees according to their learning goal orientation tendencies and managing them accordingly can minimize employee withdrawal behavior at work.

6.3. Limitations and directions for future research

We would like to point out the limitations of the present research and provide some new insights for future studies. First, the industrial adoption of digital technologies is becoming increasingly prevalent, and corporate digital responsibility has been defined as "a set of shared values and norms guiding an organization's operations with respect to the creation and operation of digital technology and data" (Lobschat et al., 2021, p. 877). Scholars have called for more discussion regarding ethical concerns related to the adoption of digital technologies (Lobschat et al., 2021; Wirtz et al., 2023). In general, the literature on service robots (including this study) has examined the phenomenon from a micro perspective, such as employees'/customers' reactions to service robots, but it is urgent and worthwhile to examine this issue from a macro perspective. For example, a CEO's moral attentiveness (i.e., the extent to which people identify or recall the moral content in an environment) (Reynolds, 2008) could be a crucial attribute in determining the extent to which a company is willing to sacrifice profit for ethical norms, and how much effort a company will put into creating a digitally responsible culture. In addition, while consumers' collective awareness of digital security could be a powerful force influencing companies' actions, future research can investigate how we can make consumers more aware of digital security.

Second, further research should also consider other boundary conditions, in addition to goal-oriented learning, to further enhance our understanding of this topic. On the individual level, some characteristics may affect how employees interact with robots and how they react to robot risk awareness. For instance, different generations may share different attitudes and worldviews on automation (Jung et al., 2021), and different levels of automation knowledge may also affect how employees perceive the innovativeness of automation. At the supervisory and organizational levels, several studies have showed that supervisor support or hotel segmentation may affect the relationships between employees' attitudes and behavioral intentions (e.g., Li et al., 2019). Future research should include moderating effects at the team and organization levels to enrich the findings.

Third, we should note that service robots and AI technologies are evolving rapidly, and future research will be needed when new generations of service robots appear in the hotel industry. Additionally, future studies should address employees' perceptions of the evolving impact of automation and the needs of sensitive employee segments that may face a greater risk of replacement.

Impact statement

Despite the rise of service robots in the tourism industry, many employees see them as a threat to their work. This paper provides important guidance on how to balance the relationship between robots and employees. By revealing the dark side of employees' perceptions of robots and exploring the causes of employees' robot risk awareness, this study proposes that hotel enterprises should consider balancing the investment ratio between robots and employees, so that employees can have a psychological balance and a clearer picture of the company's planning. The findings also suggest that a specific form of self-regulation strategy called "learning goal orientation" can enable employees to better cope with the stress and uncertainty by robots and minimize employees' withdrawal behavior at work. Therefore, our research provides managers with strategic references for utilizing employee learning goal orientation, and offers a new perspective for solving the dilemma of robots and employees' work.

CRediT authorship contribution statement

Su-Ying Pan: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Yangpeng Lin:** Writing – original draft, Visualization, Data curation. **Jose Weng Chou Wong:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization.

Declaration of competing interest

None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tourman.2024.104994.

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